Sara Alrifai

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Education

University of Waterloo | BASc, Mechatronics Engineering — Expected 2028

Internship Availability: Jan–Aug 2027 | 4 or 8 months

- **Relevant Coursework:** Machine Design, Mechanics & Dynamics, Control Systems, Systems Modeling, Sensors & Instrumentation, Actuators & Power Electronics, Microprocessors, Circuits & PCB Design, Algorithms & Data Structures

Skills

- **Mechanical Design:** SolidWorks, AutoCAD, GD&T, technical drawings, FDM 3D printing, prototyping
- **Controls & Automation:** PLCs, STM32, sensors, actuators, ADCs, equipment integration, motor control
- **Programming & Embedded:** C, C++, Python, MATLAB, SPI, Arduino, FPGA, VHDL
- **Engineering Tools:** Excel (PivotTables, VLOOKUP), Power BI, PCB/circuit design

Experience

Engineering Co-op — Sep 2025 – Apr 2026

Mondelēz International – Halls Plant — AutoCAD · PLCs · Equipment Integration · CAPEX · Power BI

- Led a production-line **PLC controls upgrade**, coordinating contractors and engineering teams through design review, equipment integration, implementation planning, and execution.
- Developed the technical scope, cost justification, and implementation strategy for a **CAPEX project**, then presented the proposal to North American governance for approval.
- Created and maintained **AutoCAD plant layouts** for equipment installations and facility modifications, including the Voltron packaging system.
- Investigated machine subassemblies through sketching, disassembly, and inspection to document **equipment operating principles** and support operator training.
- Supported the design and implementation of a **pedestrian-safety improvement project**, integrating plant layout, operational, and employee-safety requirements.
- Developed a **Power BI Gemba tracker** to monitor action items, standardize reporting, and improve visibility of outstanding plant issues.

Engineering Administrative Assistant — Jan – Apr 2025

University of Waterloo, Engineering Undergraduate Office — Excel · PivotTables · VLOOKUP · Data Analysis

- Analyzed data using **Excel, PivotTables, and VLOOKUP** for planning and reporting while coordinating engineering student documentation and event logistics.

Project Team Member, Digital Bootcamp — May – Aug 2024

WE Accelerate Program – Memotext — Agile · Responsive Design · User Testing

- Implemented responsive website features as part of an **Agile redesign** of Memotext's website, applying user-testing feedback to improve usability.
- Coordinated team tasks and virtual project meetings within an Agile workflow, contributing to milestone planning and on-time delivery.

CAD Modelling Assistant — Apr – Sep 2023

IDS Consulting — SolidWorks · AutoCAD · Technical Drawings · 3D Modelling

- Produced **AutoCAD floor plans, SolidWorks 3D models, and technical drawings** in collaboration with engineers to meet project design requirements.
- Interpreted engineering requirements and incorporated design revisions into drawings and 3D models to align deliverables with project specifications.

Projects

STM32 2-Axis Motion Control System | STM32 · C/C++ · SPI · ADC · Stepper Motors — May – Aug 2026

- Designed and programmed a **2-axis motion-control system** using an STM32F401 microcontroller and C/C++, interfacing with stepper-motor drivers through SPI.
- Implemented ADC-based speed control, GPIO, external interrupts, limit switches, and direction logic to enable controlled bidirectional motion and safe boundary detection.
- Debugged hardware-software interactions and validated **motor response, limit-switch behaviour, and boundary protection** through iterative system testing.

EV3 Prosthetic Hand | C · EV3 · Sensor Integration · Motor Control · Mechanical Design — Oct – Dec 2023

- Designed and programmed a functional **prosthetic-hand prototype** capable of controlled gripping and object manipulation.
- Integrated mechanical components, sensors, and motor-driven actuation to translate control inputs into repeatable hand motion.
- Worked on mechanical assembly, programming, hardware integration, testing, and iterative design improvements.

Certifications

WHMIS · CPR/First Aid (C) · G Driver's License